JOYCE CHEN

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EDUCATION

Rhode Island School of Design 2025 - 2027
Candidate for MID, **Master of Industrial Design**

University of Waterloo 2020 - 2025
BASc Honours **Mechatronics Engineering - Dean's Honours Distinction**

SKILLS

3D: SOLIDWORKS, OnShape, Keyshot, Creo, Blender, 3D Printing (FDM, SLA), CNC, Laser Cutting, Injection Moulding
2D: Adobe Creative Suite (Photoshop, Illustrator, Lightroom, InDesign), Figma, Procreate, Invision, AutoCAD
Programming: C, C++, MATLAB, HTML, CSS, Sass, Bootstrap, Git

EXPERIENCE

Product Design Mechanical Engineer Jun - Aug 2026, Jan - Aug 2024,
Verkada *- San Mateo, California*

- Led the complete engineering design and manufacturing of a Demo Kit device from ideation to mass production
  - Designed, kicked off, and tested complex sheet metal NCT and plastic injection moulded parts, working with manufacturing vendors in Asia for DFM and tooling
  - Worked closely with Industrial Design, Product, Electrical, Supply Chain, and Leadership to perfect the design vision and packaging
  - Created the SOP and taught assembly line workers to build over 650 units for production
- Travelled to Taiwan and Vietnam to provide mechanical engineering support on EVT and DVT builds for NPI projects
  - Supervised workers, discussed fixture/jig improvements, performed drop tests, functionality tests, cosmetic checks, fit checks
  - Unblocked unexpected issues on the line to stay on schedule

Product Design Mechanical Engineer May - Aug 2023
Viryl Technologies *- Toronto, Ontario*

- Designed and improved parts for modern automatic vinyl record pressing machines using SOLIDWORKS
- Conducted failure and root cause analysis for a keyed shaft: performed cyclic loading fatigue life calculations and proposed tolerance, material, and dimension changes to help the shaft reach infinite life
- Designed packaging to ship fragile parts to clients with suppliers, tested samples, and analyzed costs
- Performed compression moulding tests, designed parts for machining, and operated a CNC mill

Product Design Mechanical Engineer May - Aug 2022
SnapPea Design *- Kitchener, Ontario*

- Designed injection moulded parts for a wearable device and locking enclosure in Creo, used 3D printers to prototype
- Co-developed and finalized a patent pending design for a plastic clamping mechanism for fabric curtains

Mobile Game Visual Designer Sept - Dec 2021
Salutation Studio *- Waterloo, Ontario*

- Defined the artistic style and created illustrations for a mobile game with conceptual designs, splash art, typography, logos, icons, and advertisements in Figma, Adobe Creative Suite, and Procreate

UX Analyst and Web Designer Jan - Apr 2021
Ontario Institute for Cancer Research *- Toronto, Ontario*

- Co-lead the design for a new website with user scenarios, site maps, theming guides, and wireframes

PROJECTS

Redbull Flugtag 1st Place National Winner 2022

- Led the complete mechanical and artistic design of the "Snap Pea Flying Machine" in a team of 9 from creative direction and sketches to CAD and mechanical build out
- Won 1st for flight distance, creativity, and showmanship, then flew a victory lap in a Redbull stunt plane